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DATA DESCRIPTOR

A wearable sensor–based kinematic dataset collected under standardized rehabilitation tasks from 120 post-stroke patients

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Stroke frequently results in long-term motor impairments, making effective rehabilitation essential for functional recovery. However, the development of intelligent rehabilitation systems is hindered by the lack of large-scale kinematic data from stroke patients. Here, we present REHAB, a wearable sensor-based kinematic dataset collected from 120 post-stroke patients during a two-week rehabilitation program. The dataset comprises signals recorded from 27 standardized assessment movements and 16 rehabilitation training movements, providing comprehensive limb kinematics together with corresponding task annotations and clinical labels. Detailed descriptions of the data acquisition protocol, sensor configuration, and data organization are provided to ensure reproducibility and facilitate reuse. In addition, systematic quality-control procedures, including clinician-guided acquisition, sensor calibration, signal inspection, and preprocessing standardization, were implemented throughout the data collection pipeline to ensure data quality and consistency. REHAB provides a comprehensive and clinically relevant kinematic resource for stroke rehabilitation research and may support future studies in rehabilitation assessment, movement analysis, wearable sensing, and data-driven intelligent rehabilitation systems.

Background & Summary

Stroke, also known as cerebrovascular accident (CVA), is an acute disease caused by the interruption of blood flow to the brain, resulting in damage to brain tissue¹. Motor dysfunction is a common sequela among stroke patients, manifesting as limb paralysis, muscle weakness, or impaired coordination². This motor dysfunction is closely associated with injury to specific regions of the brain following a stroke, especially those responsible for regulating motor function, such as the motor cortex, basal ganglia, and spinal cord^{3,4}. To regain motor function, patients typically require long-term rehabilitation that includes professional assessments and treatments during hospitalization as well as home-based rehabilitation training post-discharge, aimed at maximizing neural plasticity and enhancing quality of life^{5–8}.

Rehabilitation assessment and training are critical for restoring motor function. Assessment serves as the foundation of rehabilitation, enabling healthcare teams to gain deep insights into patients' functional status and develop personalized rehabilitation plans^{9–11}. Standardized scales are commonly used, such as the Brunnstrom stages¹² for assessing the recovery status of hemiplegic limbs and the Fugl-Meyer Assessment (FMA)¹³ for

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assessing fine motor skills. Despite their value, clinical assessments still face challenges of being time-consuming and subjective, highlighting the need for intelligent enhancements^{14–16}. Meanwhile, home-based rehabilitation training is also crucial, especially the activity-based task-oriented training¹⁷. Its effectiveness relies heavily on patients' adherence and gradual progression¹⁸. While home programs offer the advantages of long-term exercise and increased convenience, they also present limitations. Key challenges include the lack of professional guidance, low compliance with exercise regimens, limited access to equipment and technological support, insufficient assessment mechanisms, and inadequate feedback systems^{19,20}. To address the above challenges, a widely recognized intelligent solution consists of three key steps:

Step 1: Establish a data acquisition system. Common kinematic sensors include inertial measurement units (IMUs) and cameras. IMUs can be used to measure acceleration and angular velocity to assess movement patterns and coordination²¹. Cameras can provide intuitive visual-level motion information. However, considering the convenience and privacy requirements of home-based rehabilitation, camera-based sensors are often restricted^{22,23}. In home rehabilitation, low-cost and portable wearable devices are more practical and promote broader adoption^{24,25}. Therefore, IMUs represent a practical and cost-effective solution for rehabilitation-oriented motion monitoring, providing portable, privacy-friendly, and comprehensive kinematic sensing capabilities in home-based rehabilitation scenarios^{26,27}.

Step 2: Establish data analysis models. To analyze the data collected by the acquisition system, mathematical models will be established, such as machine learning or deep learning models. By leveraging motion data, these individual models can achieve motor function assessment²⁸, rehabilitation behavior recognition²⁹, movement accuracy assessment³⁰, fatigue calculation³¹, and so on. These models can reduce workload, eliminate subjectivity, and enhance rehabilitation effectiveness.

Step 3: Build a comprehensive rehabilitation system. A comprehensive application-oriented software platform is then developed, linking the hardware and algorithms, and featuring an intuitive and visualized user interface. The platform integrates the functions of individual models, visualizes the model outputs, and provides data recording, storage, and comprehensive analysis³². Through this platform, rehabilitation progress can be continuously tracked, and detailed feedback reports can be generated. By offering real-time visual and auditory feedback, the system assists patients in correcting movements, improving adherence, and enhancing rehabilitation effectiveness.

Both the second and third steps require large volumes of pre-existing kinematic datasets for pre-training individual models, model validation, and system verification. However, existing datasets are insufficient for training models specifically designed for patients with impaired motor function, as they suffer from the following limitations:

- (1) **Lack of rehabilitation-specific tasks, limiting dataset relevance:** Existing kinematic datasets (UCI³³, Pamap2³⁴, HANDY³⁵, and³⁶) mainly contain activities of daily living (ADL), such as walking and sitting²⁹. Although they can be used to build pre-trained kinematic-based models, they lack rehabilitation-specific tasks, such as functional training exercises, fine motor movements, and resistance-based training, which are necessary to adjust the model and form a specialized one³⁷. This limitation results in insufficient diversity of rehabilitation-related movements during model training, reducing accuracy when handling complex or specialized rehabilitation tasks. Consequently, the models fail to comprehensively reflect the real needs of patients with motor impairments.
- (2) **Insufficient patient data limits model generalization and clinical applicability:** Currently, systematic kinematic datasets based on real patients suffer from insufficient data volume. For instance, IntelliRehabDS³⁸ (IRDS) contains kinematic data tailored to physical rehabilitation but is constrained by a limited number of enrolled patients. By contrast, most large-scale kinematic sensor datasets are constructed using data from healthy individuals, such as³⁹. This scarcity of patient-specific data has compelled many studies^{40,41} to rely on self-collected datasets, which typically include only healthy participants or a small cohort of patients. Consequently, models trained on such inadequate datasets fail to accurately capture the unique movement characteristics and recovery trajectories of patients. This not only impairs the models' generalization ability across diverse pathological conditions but also restricts their practical applicability in clinical settings.
- (3) **Dominance of vision-based data, limiting privacy and convenience in home rehabilitation:** Many kinematic datasets (NTU RGB + D⁴², Kinetics-Skeleton^{43,44}) primarily rely on video data, which, although effective for action recognition, raises privacy concerns in home rehabilitation settings^{45–47}. Recent advances in computer vision and single-camera body tracking frameworks have reduced the hardware and deployment requirements of vision-based systems. However, continuous video monitoring in personal living environments may still limit user acceptance and long-term applicability in home rehabilitation scenarios.
- (4) **Inconsistent labeling and low annotation quality:** To establish an intelligent model that meets clinical requirements, accurate annotation and professional labeling of the data are required. Lack of rehabilitation-specific clinical labels, such as Brunnstrom stages, Fugl-Meyer scores, or fatigue levels, will hinder the model's ability to assess rehabilitation outcomes and predict recovery stages.

To address the aforementioned limitations, we present REHAB, a wearable sensor-based kinematic dataset designed for stroke rehabilitation research. Compared with existing datasets, REHAB is characterized by a streamlined data acquisition setup, clinically grounded data collection, and comprehensive coverage of rehabilitation-related movements. In contrast to many previous IMU-based datasets that were primarily collected from healthy participants performing activities of daily living (ADLs), REHAB specifically focuses on post-stroke patients with hemiplegic motor impairments and rehabilitation-oriented movement tasks. All data in the dataset were collected from stroke patients undergoing real clinical rehabilitation procedures,

thereby providing movement characteristics that more accurately reflect pathological motor patterns and rehabilitation needs in clinical practice. Furthermore, unlike conventional ADL-oriented datasets, the movements included in REHAB are entirely composed of clinically relevant rehabilitation assessment and rehabilitation training tasks, making the dataset more suitable for rehabilitation assessment, motor function analysis, recovery monitoring, and intelligent rehabilitation applications. The dataset was acquired using a deployable and easy-to-operate sensing system composed of five wearable sensor nodes, integrating inertial measurement units (IMUs) and flex sensors (FSs). This configuration enables the collection of sufficient kinematic information while maintaining a low hardware burden, facilitating potential deployment in both clinical and home-based rehabilitation settings. REHAB comprises real-world data collected from a randomized controlled clinical trial involving 120 post-stroke patients with motor impairments over a two-week rehabilitation program. All recordings were obtained in clinical environments and accompanied by standardized clinical labels, ensuring data reliability and clinical relevance. The dataset includes kinematic recordings from 27 standardized rehabilitation assessment movements and 16 rehabilitation training movements, covering a broad range of clinically relevant motor functions. While a subset of the assessment-related kinematic data has been utilized in our previous study^{48,49} for intelligent rehabilitation assessment, these data have not been publicly released. In contrast, the kinematic data corresponding to the rehabilitation training movements are reported here for the first time. By integrating data from both rehabilitation assessment and training scenarios, REHAB provides a unified resource for studying motor assessment, recovery progression, and rehabilitation outcome analysis.

The remainder of this Data Descriptor is organized as follows. The Methods section details the clinical trial design, sensor system configuration, experimental protocol, and data preprocessing procedures. The Data Records section describes the dataset organization, data formats, and variable definitions. The Technical Validation section evaluates data quality and reliability through systematic validation analyses.

Methods

Data acquisition system. The wearable system is designed based on two key principles: (1) ease of use for donning, doffing, and operation to facilitate clinical implementation while protecting patient privacy, and (2) the ability to capture sufficient information to support intelligent assessment and training. To complete data acquisition, we utilized wearable device components including nine-axis motion sensors, rehabilitation gloves, and a Zigbee wireless receiver (Xi'an Libang Contmedu Medical Technology Co., Ltd., Xi'an, China; Remote Rehabilitation Training and Assessment System: KMD-T001; <https://www.jcyping.com/content/YiPong.action?contentID=1062&ccID=43>). These devices were tested by third-party inspectors and met the requirements for clinical research in terms of signal quality, electromagnetic compatibility (EMC), and safety. Accordingly, they have obtained a medical device license. In terms of measurement accuracy, the accuracy of angle, acceleration, and angular velocity measurements all exceeded 95%.

The data acquisition system is illustrated in Fig. 1, referred to as the Body Sensor Network (BSN). The BSN comprises four nine-axis motion sensors (S1–S4), one rehabilitation glove (S5), a ZigBee wireless receiver, and a personal host. The core of S1–S4 is nine-axis IMUs (built around the MPU9250 chip; InvenSense, USA), which simultaneously collect tri-axial acceleration, angular velocity, and magnetic field intensity signals⁵⁰. These IMUs feature a 16-bit accelerometer (adjustable range, set to ± 2 g for enhanced resolution in low-speed rehabilitation motions), a 16-bit gyroscope ($\pm 250^\circ/\text{s}$ range for optimized angular velocity accuracy), and a 14-bit magnetometer ($\pm 4800 \mu\text{T}$ range, aiding limb posture measurement and cumulative error correction). The rehabilitation glove (S5) integrates piezoresistive flex sensors (0–180° bending range) that detect hand motion by resistance changes (pressure increases with bending). An IMU module (parameters/configurations identical to S1–S4) is embedded in its back to track wrist angles. In total, S5 provides two data types: finger flexion signals (from flex sensors) and wrist flexion-extension angles (from the embedded IMU).

The sensor sampling rate is 50 Hz, which is considered sufficient for the intelligent rehabilitation field^{41,51–53}. Although minor fluctuations in the actual sampling frequency may occur during wireless data acquisition, these variations were minimal and did not noticeably affect the signal quality or subsequent analysis results. Data are transmitted wirelessly via the ZigBee protocol⁵⁴, which offers high fault tolerance, low cost, and easy networking—making it well-suited for wearable wireless motion data acquisition systems. The ZigBee receiver continuously collects sensor signals and stores the data on a personal computer, which is equipped with data acquisition software to receive and save data through connected data protocols. Sensors were worn on the affected side of the body. The rehabilitation watches were attached to the upper arm (S2), forearm (S1), thigh (S4), and lower leg (S3). Specifically, S2 was positioned approximately 8 cm above the lateral epicondyle of the humerus, while S1 was positioned approximately 10 cm above the center of the dorsal wrist crease. The S3 was positioned approximately 10–20 cm below the patella, and S4 was attached over the rectus femoris region. S5 was worn directly on the affected hand to collect finger flexion information. During data acquisition, S1, S2, and S5 were activated for upper-limb movements, whereas S3, S4, and S5 were activated for lower-limb movements.

Data acquisition protocol. *Subject and ethics statement.* The data were collected as part of a randomized controlled clinical trial. This trial recruited stroke patients with limb motor dysfunction from Tangdu Hospital of Air Force Medical University and Xi'an Gaoxin Hospital to participate in the clinical trial. This study was approved by the Ethics Committees of both Tangdu Hospital (approval No. 201912-08) and Xi'an Gaoxin Hospital (approval No. 2020-001). Written informed consent for participation and publication of de-identified data was obtained from all participants or their legal guardians. The title of the clinical trial is “Clinical trial of telerehabilitation training and intelligent evaluation systems”. The registration number is ChiCTR2200061310,

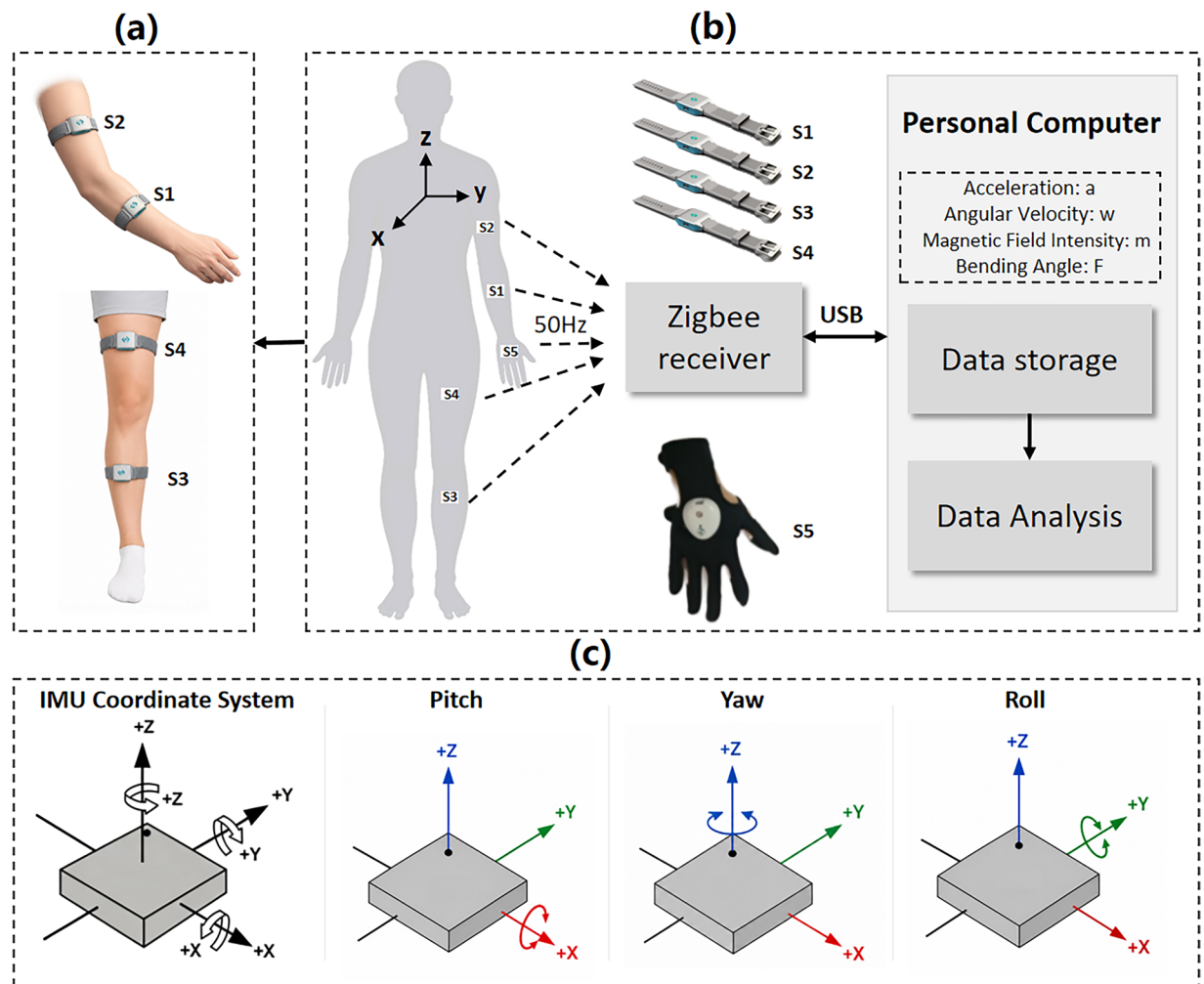


Fig. 1 Overview of the proposed data acquisition system. (a) The body sensor network (BSN) consists of four nine-axis inertial motion sensors (S1–S4), one rehabilitation glove (S5), a ZigBee wireless receiver, and a personal host computer. Sensors S1–S4 are based on nine-axis IMUs (MPU9250) that acquire tri-axial acceleration, angular velocity, and magnetic field signals. The rehabilitation glove integrates piezoresistive flex sensors to measure finger flexion and an embedded IMU to capture wrist motion. All sensors operate at an approximate sampling frequency of 50 Hz, and data are transmitted wirelessly via the ZigBee protocol to the host computer for storage. (b) Sensor placement configuration. Sensors were worn on the affected side of the body: forearm (S1), upper arm (S2), lower leg (S3), thigh (S4), and hand (S5). Specifically, S2 was positioned approximately 8 cm above the lateral epicondyle of the humerus, while S1 was positioned approximately 10 cm above the center of the dorsal wrist crease. S3 was positioned approximately 10–20 cm below the patella, and S4 was attached over the rectus femoris region. Sensors S1, S2, and S5 were activated for upper-limb data acquisition, whereas S3, S4, and S5 were activated for lower-limb movements. (c) Standardized sensor coordinate system. Before each recording session, the IMU coordinate system was automatically calibrated. In the standardized sensor coordinate system, the X-axis was defined as perpendicular to the coronal plane, the Y-axis as perpendicular to the sagittal plane, and the Z-axis as perpendicular to the transverse plane.

registered on 2022-06-20 in the Chinese Clinical Trial Registry. All methods were conducted in accordance with the Declaration of Helsinki and relevant institutional and national guidelines and regulations.

Inclusion criteria. (1) Stroke patients diagnosed by Computed Tomography (CT) or Magnetic Resonance Imaging (MRI); (2) Aged between 30 and 75 years, with no gender restrictions; (3) Post-stroke duration of 15–180 days with limb motor dysfunction (subacute post-stroke patients); (4) Brunnstrom stage II–VI in the upper and lower limb, with stable condition; (5) Normal cognitive function, able to understand and comply with the study protocol; (6) Good compliance with the study requirements.

Exclusion criteria. (1) Severe cognitive impairment that prevents completion of FMA; (2) Presence of fractures or other significant limb injuries; (3) Joint contracture formation; (4) Patients classified as disabled according to the International Classification of Functioning, Disability and Health (ICF) standards; (5) Presence of severe

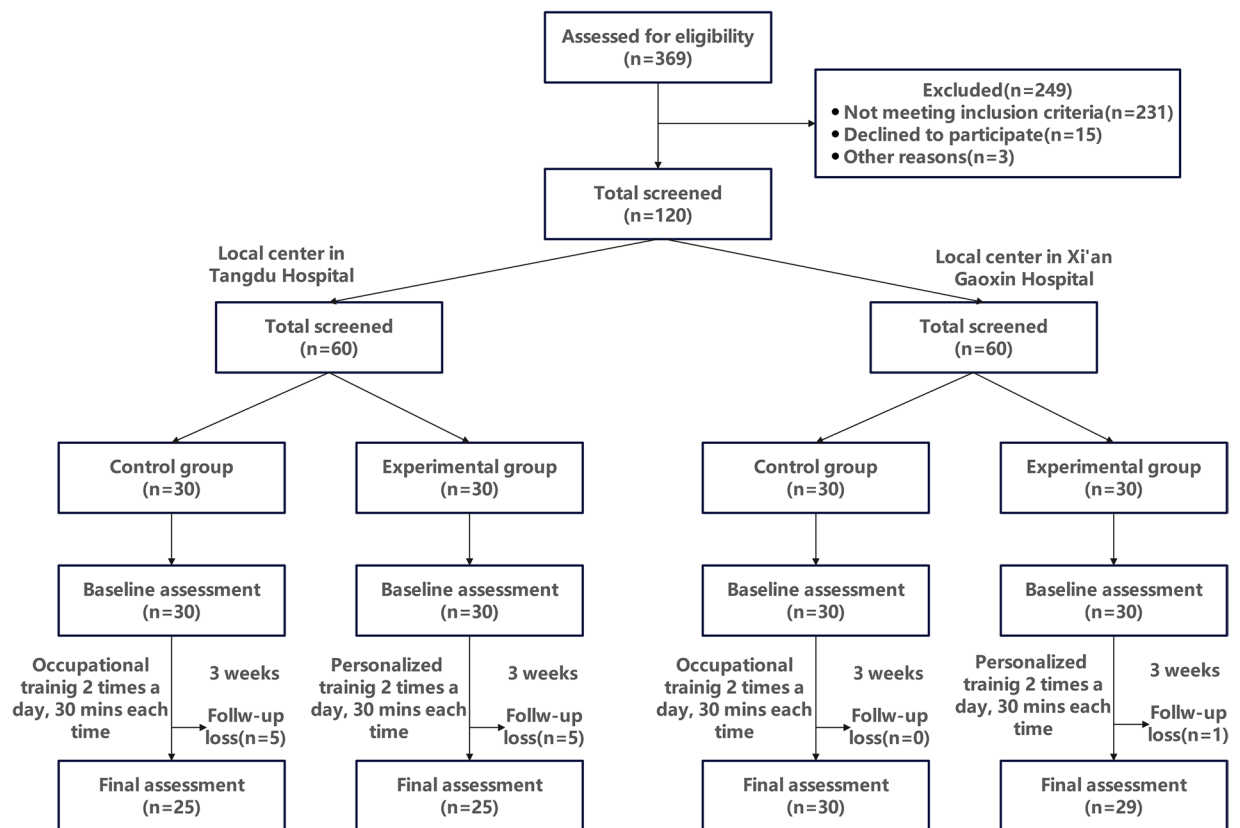


Fig. 2 Clinical trial workflow. A total of 369 participants were screened across two medical centers, of whom 249 were excluded. The remaining 120 eligible participants were randomly assigned to experimental and control groups (60 per center). During the trial, 11 participants withdrew, and 109 participants completed the study.

primary diseases; (6) Positive urine pregnancy test in women of childbearing age; (7) Participation in other drug/device clinical trials within the past 90 days; (8) Implanted active metallic devices (e.g., pacemakers); (9) Any other conditions deemed unsuitable for the study by the investigators.

The clinical trial workflow is illustrated in Fig. 2. A total of 369 participants were screened for eligibility across two medical centers. Among them, 249 were excluded (231 did not meet the inclusion criteria, 15 declined participation, and 3 were excluded for other reasons). A total of 120 eligible participants were enrolled from two clinical centers and randomly assigned to either the experimental group or the control group. Tangdu Hospital independently recruited 60 participants, who were randomly allocated into the experimental group ($n = 30$) and control group ($n = 30$) using a computer-generated randomization sequence. Similarly, Xi'an Gaoxin Hospital independently recruited 60 participants and randomly assigned them into the experimental group ($n = 30$) and control group ($n = 30$) following the same randomization procedure. However, during the actual clinical trial, based on the principle of voluntary participation, 10 participants from Tangdu Hospital and 1 participant from Xi'an Gaoxin Hospital withdrew, resulting in a total dropout of 11 participants. The reasons for withdrawal included physical discomfort during the rehabilitation process, worsening of medical conditions, and unwillingness to continue participation in the trial. In the end, 109 participants completed the entire clinical trial, achieving a completion rate of 91%. The information of the 120 enrolled participants at baseline is presented in Table 1, with additional detailed information provided in supplementary Table 1.

The trial lasted 2 weeks. At baseline, both the control and experimental groups (120 participants in total) underwent baseline rehabilitation assessments. During the 2-week intervention period:

The control group received two types of conventional training. They underwent physical therapy (PT) training twice daily, with each session lasting 30 minutes, and completed conventional occupational therapy (OT) training twice daily. This training schedule ensured a minimum of 10 training sessions per week and a total of at least 30 sessions over the entire intervention period.

The experimental group received a combination of personalized and conventional training. They followed a personalized training protocol (two 30-minute sessions per day) and completed conventional OT training twice daily. This personalized protocol involved selecting appropriate training movements and durations from a pre-defined movement library, tailored to each patient's specific motor impairments by a physician.

At the end of the 2-week trial, all remaining participants in both groups (109 participants in total) underwent a final rehabilitation assessment. All rehabilitation training sessions and assessments conducted throughout the

Characteristics	Control group (n = 60)			Experimental group (n = 60)			
	Values ^a	95% CI	Range(median)	Values ^a	95% CI	Range(median)	P value
Age (year)	55.82 (9.68)	53.32–58.32	34.00–73.00 (55.00)	56.25 (9.83)	53.71–58.79	33.00–73.00 (56.00)	0.81
Gender (male)	43 (71.67)	—	—	43 (71.67)	—	—	>0.99
Course of disease (days)	61.20 (46.67)	49.15–73.25	15.00–169.00 (41.00)	46.22 (36.63)	36.75–55.68	15.00–165.00 (33.00)	0.15
Systolic blood pressure (SBP)(mmHg)	127.47 (11.56)	124.48–130.45	96.00–151.00 (127.00)	130.13 (13.73)	126.59–133.68	92.00–166.00 (130.00)	0.25
Diastolic blood pressure (DBP) (mmHg)	81.05 (9.00)	78.73–83.37	60.00–105.00 (80.00)	82.45 (9.51)	79.99–84.91	57.00–107.00 (80.50)	0.41
Body Mass Index (BMI)(kg/m ²)	24.68 (4.08)	23.59–25.76	18.34–44.92 (23.70)	24.66 (2.59)	23.99–25.33	18.34–44.92 (24.57)	0.39
Stroke type (cerebral infarction)	37 (61.67)	—	—	32 (53.33)	—	—	0.36
Hypertension (HTN)	44 (73.33)	—	—	43 (71.67)	—	—	0.84
Hyperlipidemia (HLP)	7 (11.67)	—	—	4 (6.67)	—	—	0.53
Arteriosclerotic coronary disease / myocardial infarction (ACD/MI)	10 (16.67)	—	—	12 (20.00)	—	—	0.64

Table 1. The detailed information of the 120 enrolled participants at baseline. Values^a; data are mean (SD) or n (%). Confidence Interval (CI).

Flexor Synergy Movement	0	Shoulder elevation
	1	Shoulder retraction
	2	Shoulder abduction (at least 90°)
	3	Shoulder external rotation
	4	Elbow flexion
	5	Forearm supination
Extensor Synergy Movement	6	Shoulder adduction and internal rotation
	7	Elbow extension
	8	Forearm pronation
Upper Limb Synergistic Movement	9	Hand touching the lumbar spine
	10	Shoulder flexion to 90° (with elbow joint at 0°)
	11	Forearm pronation or supination (with shoulder joint at 0° and elbow joint at 90°)
Isolated Movement	12	Forearm pronation (with shoulder abduction at 90° and elbow joint at 0°)
	13	Forearm pronation and supination (with shoulder flexion ranging from 90 to 180° and elbow joint at 0°)
	14	Forearm pronation or supination (with shoulder flexion between 30 to 90° and elbow joint at 0°)
Wrist Movement	15	Wrist dorsiflexion (with elbow joint at 90° and shoulder joint at 0°)
	16	Wrist flexion and extension (with elbow joint at 90° and shoulder joint at 0°)
	17	Wrist dorsiflexion (with elbow joint at 0° and shoulder joint at 30°)
	18	Wrist flexion and extension (with elbow joint at 0° and shoulder joint at 30°)
	19	Wrist circumduction
Hand Fine Motor Movement	20	Finger flexion (flexion of all fingers together)
	21	Finger extension (extension of all fingers together)
	22	Thumb adduction with all joints at 0° position
	23	Thumb pinch (holding a pencil with the thumb)
	24	Grasp a cylindrical object
	25	Grasp a spherical object
Complex Coordinated Movement	26	Finger-to-nose test (performed five times consecutively)

Table 2. Rehabilitation assessment items selected from the FMA, consisting of 27 items.

trial were closely monitored by professional rehabilitation therapists, who provided assistance to participants when necessary to prevent injuries during the trial.

Rehabilitation assessment data collection. These data were collected from the baseline rehabilitation assessment (involving 120 patients) and the final rehabilitation assessment at discharge (involving 109 patients). The rehabilitation assessment movements were selected from the Fugl-Meyer Assessment for the Upper Extremity (FMA-UE). Although the original FMA-UE contains 33 items, six items were excluded in this study for clinical and technical reasons, resulting in 27 retained assessment items, as shown in Table 2. Specifically, three reflex-related items were excluded because all enrolled patients were at Brunnstrom Stages II–VI and already exhibited reflex activity, resulting in limited discriminative information for these items. In addition, the grip strength assessment item was excluded because quantitative acquisition of grip force would require

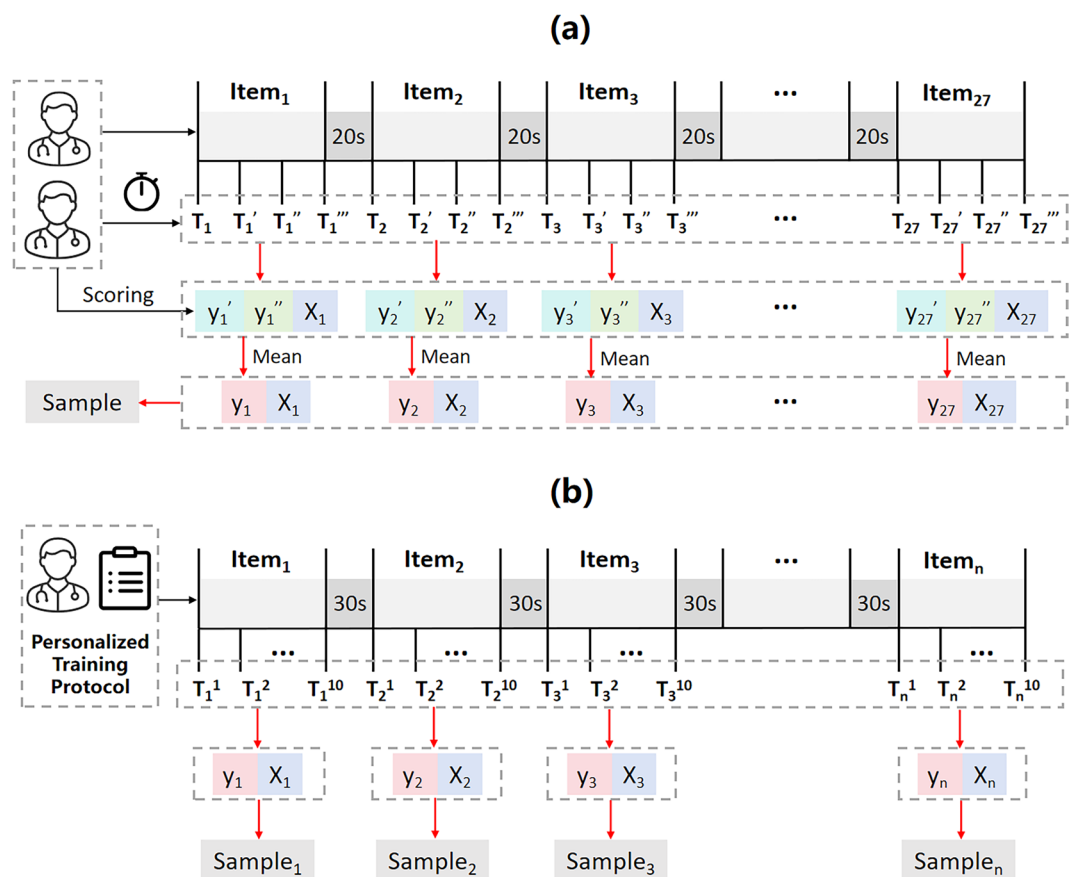


Fig. 3 Data acquisition procedures. **(a)** Rehabilitation assessment data collection was jointly conducted by two rehabilitation therapists. A total of 27 assessment items were collected, with each item repeated 3 times. One therapist guided the patient to perform the assessment movements sequentially, with a 20-s rest interval between items, while the other therapist recorded key time points using a timer. The two therapists independently scored each item, producing scores y' and y'' ; the final score label y was obtained by taking the ceiling of their mean value. The kinematic data of the 27 assessment items (X_1 - X_{27}) together with their corresponding 27 score labels (y_1 - y_{27}) were defined as one sample. **(b)** Rehabilitation training data collection was conducted by a single therapist. The training items were selected from each patient's personalized rehabilitation training protocol. Each item was repeated approximately 10 times, and all prescribed items were performed sequentially with a 30-s rest interval between items. For each training item, the collected kinematic data X_i together with its corresponding movement label y_i were defined as a single sample.

dedicated force-sensing hardware that was not included in the current wearable sensing system. Furthermore, two coordination-related items were removed because they represented highly similar movement patterns and would introduce redundancy into the dataset. The data acquisition procedure for a single subject is illustrated in Fig. 3(a). Before the assessment, a clinician configured the BSN system, calibrated the IMU sensors, assisted the patient in wearing the sensors, and verified their correct placement to ensure reliable data acquisition. During the assessment, one therapist guided the participants through all items in sequence, with each item repeated three times and a 20-second rest interval between items, while a second therapist annotated key time points. During the rest intervals, both therapists independently scored each item according to the FMA-UE criteria to serve as clinical labels. A score of 0 indicates no movement or completely uncoordinated movement; a score of 1 indicates partial completion of the movement; and a score of 2 indicates perfect execution of the movement. The final rating label was determined as the ceiling of the average score provided independently by the two therapists. As the purpose of this phase was to assess motor function rather than training, each movement was repeated only three times, which was sufficient for clinicians to evaluate the corresponding motor function. Throughout the assessment, wearable sensors continuously captured the patients' motion signals in real time. The collected data were wirelessly transmitted to a computer via the ZigBee protocol and stored for subsequent analysis.

In total, we completed 120 + 109 rehabilitation assessments, with each assessment corresponding to one data collection session. Each data collection session included motion data from the 27 assessment items, where the data for each item is denoted as $X_i \in \mathbb{R}^{N \times C}$, $i \in \{1, 2, 3, \dots, 27\}$. Here, C represents the number of signal channels, and N denotes the signal length. Due to variations in the speed at which patients performed the movements, the signal length N varied across patients; this variation will be addressed by standardizing the signal length in the data preprocessing section. We define a "sample" as all data collected from a single assessment, denoted as: sample = $\{(X_1, y_1), (X_2, y_2), \dots, (X_{27}, y_{27})\}$, where $X_i \in \mathbb{R}^{N \times C}$ represents the signal data and $y_i \in \{0, 1, 2\}$ denotes the scoring label.

Upper extremity combined training (Arm/Fingers/Wrist/Shoulder)	0	bobath handshake
	1	bobath flexion/extension
	2	bobath forward flexion/extension
	3	bobath anterior/posterior rotation
	4	elbow flexion and wrist compression
Hands training (Fingers/Wrist)	5	wrist flexion and extension
	6	finger-to-finger training
	7	ball gripping
Shoulder training (Shoulder)	8	shoulder joint internal and external
	9	breast expansion
Arm training (Arm)	10	flexion-pressure rotation forward and backward
	11	elbow joint flexion and touch
	12	shoulder touch training
Lower limbs training (Ankle/Knee/Hip)	13	Ankle extension & Knee internal and external rotation
	14	knee flexion and extension
	15	Hip flexion and extension

Table 3. Rehabilitation training movement library, consisting of 16 items.

Rehabilitation training data collection. The kinematic data of rehabilitation training were collected when the experimental group underwent personalized training during the trial period. The training movement library consists of 16 predefined items^{55–58}, as listed in Table 3. These 16 items target key regions of the upper limbs, including the shoulder, elbow, and arm, as well as the lower limbs, including the hip, thigh, and knee joint. In addition to conventional Occupational Therapy (OT) training, patients in the control group received training according to personalized training protocols. These protocols were developed by rehabilitation clinicians, who selected a subset of items from a predefined movement library based on each patient's specific condition and motor abilities. The data acquisition procedure for a single subject is illustrated in Fig. 3(b). Before the assessment, a clinician configured the BSN system, calibrated the IMU sensors, assisted the patient in wearing the sensors, and verified their correct placement to ensure reliable data acquisition. Once confirmed, the patient performed training in accordance with the protocol. Each item in the personalized training protocol was performed approximately 10 times. This repetition count was selected to ensure participants achieved the required training effect, which helps promote muscle activation and endurance. After completing one item, participants were given a 30-second rest period before moving on to the next item; this process continued until all items in the protocol were finished, and the entire training session lasted approximately 20–30 minutes. The wearable devices continuously capture patients' motion data in real time and transmit it via the ZigBee protocol.

In total, 4,689 data entries were collected from 60 patients over the two-week training period, covering all 16 training items. Each data entry is denoted as $X_i \in \mathbb{R}^{N \times C}$ (where $i = 1, 2, 3, \dots, 16$), with C representing the number of signal channels and N denoting the signal length. Due to variations in the speed at which patients performed the movements, the signal length N differed across data entries; this variation will be addressed by standardizing the signal length in the data preprocessing section. We define a “sample” as data from a full motion cycle of a certain movement performed by a patient during a single training session, denoted as: sample = (X_i, y_i) , where $X_i \in \mathbb{R}^{N \times C}$ represents the signal data and $y_i \in \{0, 1, \dots, 16\}$ denotes the movement type. Since each patient in the experimental group followed a distinct personalized training protocol—and there were also slight variations in the frequency of training sessions among patients—the number of samples collected for each of the 16 training items varied accordingly.

Data processing. Euler angle orientation estimation. For motor function assessment or recognition, acceleration and angular velocity signals are not intuitive for direct interpretation. In contrast, posture parameters, such as Euler angles, offer a clearer and more meaningful representation of movement patterns. Euler angles use three rotational angles to define the orientation of an object in three-dimensional space: Yaw represents rotation about the vertical axis, Pitch represents rotation about the horizontal transverse axis, and Roll represents rotation about the longitudinal axis, expressed as: $EA = \{\text{Yaw}, \text{Pitch}, \text{Roll}\}$. To extract posture information, we propose an orientation estimation algorithm based on quaternion calculation^{59,60}, which converts the nine-axis IMU signals into Euler angles, providing a direct and interpretable description of motion orientation.

The quaternion q is constructed as Eq. (1):

$$q = m + xi + yj + zk \quad i^2 + j^2 + k^2 = -1 \quad (1)$$

where m is the real part, and x, y , and z are the imaginary parts.

Then, by using the angular velocity data, the increment of the quaternion Δq can be calculated via Eq. (2):

$$\Delta q = \left(\cos\left(\frac{\|w\|\Delta t}{2}\right), \sin\left(\frac{\|w\|\Delta t}{2}\right) \cdot \frac{w_x}{\|w\|}, \sin\left(\frac{\|w\|\Delta t}{2}\right) \cdot \frac{w_y}{\|w\|}, \sin\left(\frac{\|w\|\Delta t}{2}\right) \cdot \frac{w_z}{\|w\|} \right) \quad (2)$$

where $\|w\|$ is the magnitude of the angular velocity.

Next, Δq is used to update the quaternion:

$$q_{\text{gyro}} = m' + x'i + y'j + z'k = q \otimes \Delta q \quad (3)$$

where $\otimes$ denotes quaternion multiplication; q_{gyro} is the updated quaternion; m' is the updated real part, and x' , y' , and z' are the updated imaginary parts.

Then, the quaternion can be converted into Euler angles via Eqs. (4–6).

$$\text{roll} = \arctan2(2(y'z' + w'x'), (w')^2 + (x')^2 - (y')^2 - (z')^2) \quad (4)$$

$$\text{pitch} = \arcsin(2(y'w' - x'z')) \quad (5)$$

$$\text{yaw} = \arctan2(2(x'y' + z'w'), (w')^2 - (x')^2 - (y')^2 + (z')^2) \quad (6)$$

Signal Segmentation and Interpolation. For the rehabilitation assessment data, movement segmentation was performed by aligning the manually recorded start–end times of each assessment task with the timestamp information contained in the continuously recorded sensor signals. Each assessment movement was repeated three times, and the second repetition was selected as the representative motion cycle for further analysis. After segmentation, only the selected motion cycle signal was retained, while the remaining repetitions were discarded. Subsequently, linear interpolation was applied to standardize the length of the single motion cycle signal to 150 data points. Specifically, for each X_i in a sample, we first conducted manual signal segmentation, followed by linear interpolation to standardize the signal length. After the above processing, the sample was updated to “sample = $\{(X_1, y_1), (X_2, y_2), \dots, (X_i, y_i), \dots, (X_{27}, y_{27})\}$ ”, where $X_i \in \mathbb{R}^{150 \times C}$, $y_i \in \{0, 1, 2\}$, and X_i contained only one motion cycle.

For the rehabilitation training data, variations in movement duration and cycles among patients led to significant differences in signal length. To standardize the data while preserving its original characteristics, the signals were truncated or padded to a fixed length of 880 data points: If the original signal length exceeded 880, only the first 880 data points were retained; If the original signal length was shorter than 880, zero-padding was applied at the end of the signal to meet the fixed length requirement. After the above processing, each rehabilitation training motion sample X_i satisfied $X_i \in \mathbb{R}^{880 \times C}$, and each X_i contained 8–10 motion cycles.

Filtering and Normalization. To eliminate signal instability and spikes, a moving average filter⁶¹ with a window length of 10 was applied. Baseline drift will cause “useless offset information” in the data to obscure the effective fluctuations of the features themselves (such as minor changes reflecting the core state in experimental signals). The neural network may mistakenly learn the baseline drift as a “significant feature” instead of focusing on the real signal variation patterns, ultimately leading the model to learn noise or irrelevant information and reducing the prediction accuracy for the target task. To reduce baseline drift caused by individual differences and ensure consistency across samples, all signals underwent zero-mean normalization⁶², ensuring that each signal had a mean of zero and a consistent variance across the dataset.

Error sample processing. During the experiment, factors such as equipment malfunctions, human errors, and other issues led to data loss and missing samples. To address this, we manually identified and removed these erroneous samples. After filtering, the number of assessment samples was reduced from 229 to 222, and the number of training samples decreased from 4,689 to 4,616.

Data Records

We reorganized the data structure and uploaded the dataset to Science Data Bank (Science DB)⁶³. The main dataset file is named **REHAB** (level-1), and its structure is shown in Fig. 4. The main file contains two subfolders: **Rehab_assessment** (level-2) and **Rehab_exercise** (level-2) (“level-1”, “level-2”, and “level-3” refer to the hierarchical folder organization structure of the dataset).

Rehab_assessment structure. **Rehab_assessment** (level-2) contains two subfolders, **d01_raw_data** (level-3) and **d02_processed_data** (level-3), along with a label file **Label.npy**.

d01_raw_data (level-3): Contains the raw data from 222 Fugl-Meyer assessments from 120 patients. It includes 222 subfolders named 001–222 (level-4), each corresponding to a unique assessment sample (sample001–sample222). Within each sample folder (e.g., 001), data of the 27 movements are stored as **.npy** files, each representing a single movement–sensor pair. All files are saved in NumPy array format and can be loaded using **numpy.load()**.

Naming Convention: Each **.npy** file follows the pattern **<movementID>_<sensorID>.npy** with dimensions: (signal_length $\times$ number_of_channels), where **<movementID>** (000–026) refers to the movements listed in Table 2, and **<sensorID>** (1,2,3) specifies the sensor configuration included in that movement, as defined in Table 4.

When **<sensorID> = 1**, it contains 12 channels: Pitch₁, yaw₁, roll₁, ax₁, ay₁, az₁, mx₁, my₁, mz₁, wx₁, wy₁, and wz₁ (see Table 5). The corresponding data tensor has dimensions **<movementID>_1 $\in \mathbb{R}^{N \times 12}$** , where N denotes

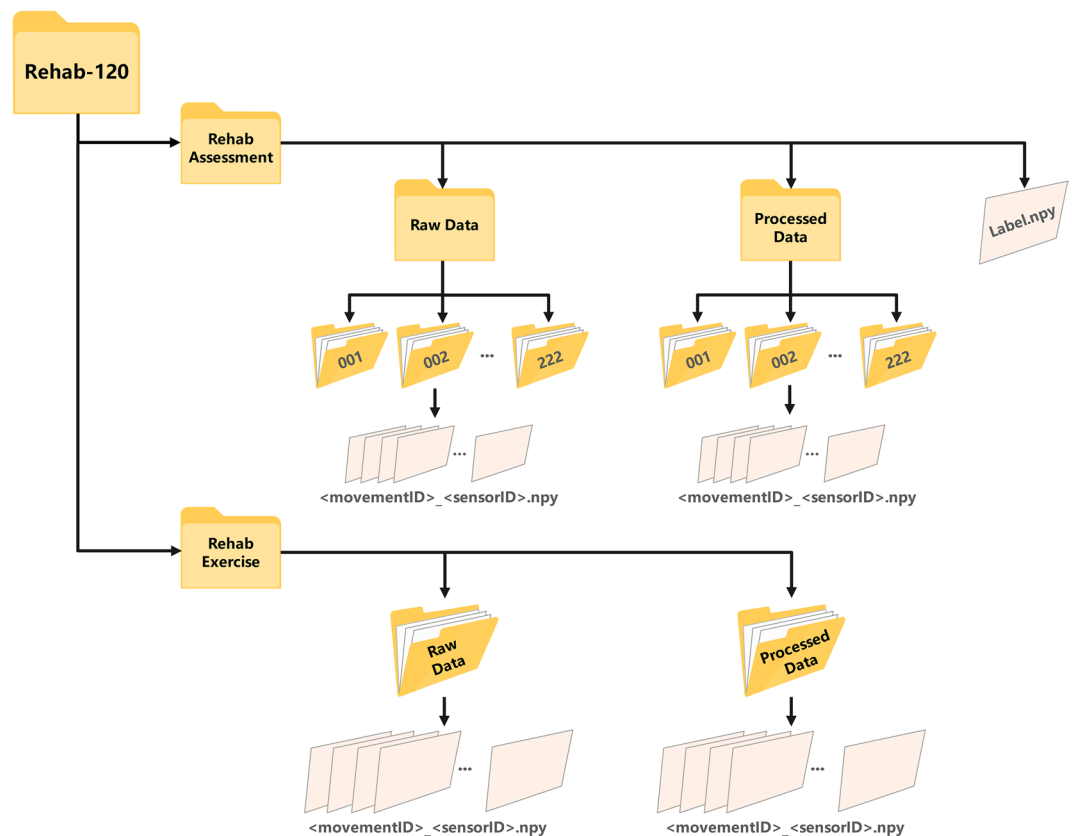


Fig. 4 Dataset file organization. The dataset consists of two subdirectories. The first, **Rehab_assessment**, contains rehabilitation assessment motion data with corresponding labels. The second, **Rehab_exercise**, contains rehabilitation training motion data.

<sensorID>	Represented Sensors	Channels (C)
1	S1: When the movement is an upper limb movement; S3: When the movement is a lower limb movement.	12
2	S2: When the movement is an upper limb movement; S4: When the movement is a lower limb movement.	13
3	S5	5

Table 4. Sensor configurations for rehabilitation assessment data corresponding to different <sensorID> values.

the signal length. For upper-limb movements, this corresponds to S1, whereas for lower-limb movements, it corresponds to S3. For example, “001_1.npz” represents the signal data from S1 for movement 1.

When <sensorID> = 2, it contains 13 channels: pitch₂-pitch₁, pitch₂, yaw₂, roll₂, ax₂, ay₂, az₂, mx₂, my₂, mz₂, wx₂, wy₂, and wz₂ (see Table 6). The corresponding data tensor has dimensions <movementID>_2 ∈ ℝ^{N×13}, where N denotes the signal length. For upper-limb movements, this corresponds to S2, whereas for lower-limb movements, it corresponds to S4. For example, “001_2.npz” represents the signal data from S2 for movement 1.

When <sensorID> = 3, the signal contains 5 channels: pitch₃, pitch₃-pitch₁, f₁, f₁₂, and f_{all} (see Table 7). The corresponding data tensor has dimensions <movementID>_3 ∈ ℝ^{N×5}, where N denotes the signal length. This corresponds to S5. For example, “001_3.npz” represents the signal data from S5 for movement 1.

d02_processed_data (level-3): The same structure as **d01_raw_data**, but with preprocessed data. The pre-processing steps include: Moving average filtering, Zero-mean normalization, and Linear interpolation. After data preprocessing, the signal length was fixed at N = 150.

Label.npz: A NumPy array of shape 222 × 27, where 222 corresponds to the number of assessment samples and 27 corresponds to the movements listed in Table 2. This file contains the scoring labels for all 27 movements across the 222 samples, as evaluated by rehabilitation therapists.

Rehab_exercise structure. **Rehab_exercise** (level-2) contains two subfolders, **d01_raw_data** (level-3) and **d02_processed_data** (level-3).

Header name	Header Description	Units of Measurement
pitch ₁	The angle of rotation around the X-axis, representing up-downward tilting of S1 or S3.	degrees (°)
yaw ₁	The angle of rotation around the Z-axis, representing left-right turning of S1 or S3.	degrees (°)
roll ₁	The angle of rotation around the Y-axis, representing tilting or rolling of S1 or S3.	degrees (°)
ax ₁	Acceleration component in the X-axis direction.	m/s ²
ay ₁	Acceleration component in the Y-axis direction.	m/s ²
az ₁	Acceleration component in the Z-axis direction.	m/s ²
mx ₁	Magnetic field strength component in the X-axis direction.	T
my ₁	Magnetic field strength component in the Y-axis direction.	T
mz ₁	Magnetic field strength component in the Z-axis direction.	T
wx ₁	Angular velocity component around the X-axis.	rad/s
wy ₁	Angular velocity component around the Y-axis.	rad/s
wz ₁	Angular velocity component around the Z-axis.	rad/s

Table 5. Interpretation of channel headers for rehabilitation assessment data when <sensorID> = 1.

Header name	Header Description	Units of Measurement
pitch ₂ -pitch ₁	The difference between pitch ₁ and pitch ₂ .	degrees (°)
pitch ₂	The angle of rotation around the X-axis, representing up-downward tilting of S2 or S4.	degrees (°)
yaw ₂	The angle of rotation around the Z-axis, representing left-right turning of S2 or S4.	degrees (°)
roll ₂	The angle of rotation around the Y-axis, representing tilting or rolling of S2 or S4.	degrees (°)
ax ₂	Acceleration component in the X-axis direction.	m/s ²
ay ₂	Acceleration component in the Y-axis direction.	m/s ²
az ₂	Acceleration component in the Z-axis direction.	m/s ²
mx ₂	Magnetic field strength component in the X-axis direction.	T
my ₂	Magnetic field strength component in the Y-axis direction.	T
mz ₂	Magnetic field strength component in the Z-axis direction.	T
wx ₂	Angular velocity component around the X-axis.	rad/s
wy ₂	Angular velocity component around the Y-axis.	rad/s
wz ₂	Angular velocity component around the Z-axis.	rad/s

Table 6. Interpretation of channel headers for rehabilitation assessment data when <sensorID> = 2.

Header name	Header Description	Units of Measurement
pitch ₃ -pitch ₁	The difference between pitch ₃ and pitch ₁ .	degrees (°)
pitch ₃	The angle of rotation around the X-axis, representing up-downward tilting of S5.	degrees (°)
f ₁	Thumb flexion angle.	degrees (°)
f ₁₂	The sum of the flexion angles of the thumb and index finger.	degrees (°)
f _{all}	The sum of the flexion angles of the five fingers.	degrees (°)

Table 7. Interpretation of channel headers for rehabilitation assessment data when <sensorID> = 3.

d01_raw_data (level-3): contains the raw data for 16 training movements stored as **.npz** files. Each **.npz** file follows the pattern <movementID>_<sensorID>.**.npz** with dimensions: (sample_size × signal_length × number_of_channels), where <movementID> (000-015) refers to the movement category listed in Table 3, and <sensorID> (1,2) specifies the sensor configuration included in that movement, as defined in Table 8.

When <sensorID> = 1 (6 channels), it contains the 6 channels: pitch₁, yaw₁, roll₁, pitch₂, yaw₂, and roll₂ (see Table 9). The corresponding data tensor has dimensions <movementID>_1 ∈ ℝ^{d×880×6}, where d denotes the number of samples for the given movement, and 880 is the standardized signal length. For upper-limb movements, it corresponds to Sensors S1 and S2, whereas for lower-limb movements, it corresponds to Sensors S3 and S4. For example, “001_1.npz” represents the signal data from S1 and S2 for movement 1.

When <sensorID> = 2 (6 channels), it contains the 6 channels: f₁, f₂, f₃, f₄, f₅, and pitch₃ (see Table 10). The corresponding data tensor has dimensions <movementID>_2 ∈ ℝ^{d×880×6}, where d denotes the number of samples for the given movement, and 880 is the standardized signal length. It corresponds to Sensor S5. “001_2.npz” represents the signal data from S5 for movement 1.

<sensorID>	Represented Sensors	Channels (C)
1	S1 and S2: When the movement is an upper limb movement; S3 and S4: When the movement is a lower limb movement.	6
2	S5	6

Table 8. Sensor configurations for rehabilitation training data corresponding to different <sensorID> values.

Header name	Header Description	Units of Measurement
pitch ₁	The angle of rotation around the X-axis, representing up-downward tilting of S1 or S3.	degrees (°)
yaw ₁	The angle of rotation around the Z-axis, representing left-right turning of S1 or S3.	degrees (°)
roll ₁	The angle of rotation around the Y-axis, representing tilting or rolling of S1 or S3.	degrees (°)
pitch ₂	The angle of rotation around the X-axis, representing up-downward tilting of S2 or S4.	degrees (°)
yaw ₂	The angle of rotation around the Z-axis, representing left-right turning of S2 or S4.	degrees (°)
roll ₂	The angle of rotation around the Y-axis, representing tilting or rolling of S2 or S4.	degrees (°)

Table 9. Interpretation of channel headers for rehabilitation training data when <sensorID> = 1.

Header name	Header Description	Units of Measurement
f ₁	Thumb flexion angle.	degrees (°)
f ₂	Index finger flexion angle.	degrees (°)
f ₃	Middle finger flexion angle.	degrees (°)
f ₄	Ring finger flexion angle.	degrees (°)
f ₅	Little finger flexion angle.	degrees (°)
pitch ₃	The angle of rotation around the X-axis, representing up-downward tilting of S5.	degrees (°)

Table 10. Interpretation of channel headers for rehabilitation training data when <sensorID> = 2.

d02_processed_data (level-3): Follows the same structure as **d01_raw_data**, but the data has undergone preprocessing, including moving average filtering and zero-mean normalization.

Technical Validation

Data acquisition quality control. To ensure the reliability, consistency, and clinical validity of the collected data, standardized quality control procedures were implemented throughout the entire data acquisition process, including subject enrollment, sensor setup, movement execution, signal acquisition, and post-processing.

First, all rehabilitation assessments and rehabilitation training sessions were conducted in real clinical environments under the supervision of professional rehabilitation therapists. Before formal participation, all operators—including rehabilitation therapists and research staff—received unified training regarding sensor placement, system operation, movement guidance, standardized assessment procedures, and data recording workflows. This standardized training ensured consistency across clinical centers, subjects, and acquisition sessions. Before each recording session, the BSN system was systematically inspected. The inspection procedure included verification of sensor connectivity, ZigBee wireless transmission stability, battery status, signal synchronization, and communication integrity between the wearable sensors and the host computer. All rehabilitation assessment tasks were performed according to standardized Fugl-Meyer Assessment procedures, and all rehabilitation training tasks were executed following clinician-defined rehabilitation protocols. Throughout the recording process, rehabilitation therapists continuously supervised movement execution to ensure that the movements were completed according to clinical instructions and to minimize variability caused by incorrect movement patterns or compensatory actions. After data collection, all recordings underwent additional manual quality screening and preprocessing. Error samples caused by equipment malfunction, human operation errors, or missing data were manually identified and removed. Furthermore, moving average filtering, zero-mean normalization, interpolation, and signal length standardization were applied to improve signal consistency while preserving the original movement characteristics.

These quality control procedures ensured that the collected dataset maintained high reliability, consistency, and clinical relevance, while minimizing variability introduced by device defects, inconsistent sensor placement, signal transmission failures, or improper movement execution.

Validation of sensor measurement accuracy. We further conducted accuracy validation experiments for both the rehabilitation glove and the wearable IMU sensors by comparing the sensor output angles with pre-defined standard angles under multiple rehabilitation movement conditions. Specifically, a series of standard joint angles were manually set, and the corresponding angular values measured by the wearable sensors were recorded and compared with the reference values to evaluate measurement accuracy and stability.

As shown in Table 11, the rehabilitation glove exhibited angular measurement errors ranging from +1° to +3°, all within the predefined allowable error range of ±10°. Meanwhile, the wearable IMU system

Motion Type	Target Joint / Movement	Set Angle (°)	Measured Angle (°)	Error (°)	Requirement	Result
Rehabilitation Glove	Finger flexion	30	31	+1.0	±10°	Passed
Rehabilitation Glove	Finger flexion	55	58	+3.0	±10°	Passed
Rehabilitation Glove	Finger flexion	88	90	+2.0	±10°	Passed
Wearable IMU	Shoulder flexion	58.3	59	+0.7	±5°	Passed
Wearable IMU	Shoulder flexion	92.2	90	−2.2	±5°	Passed
Wearable IMU	Shoulder flexion	152.5	150	−2.5	±5°	Passed
Wearable IMU	Shoulder flexion	176.8	175	−1.8	±5°	Passed
Wearable IMU	Forearm supination	35.6	37	+1.4	±5°	Passed
Wearable IMU	Forearm supination	72.4	74	+1.6	±5°	Passed
Wearable IMU	Elbow flexion	54.6	52	−2.6	±5°	Passed
Wearable IMU	Elbow flexion	118.8	120	+1.2	±5°	Passed
Wearable IMU	Elbow flexion	126.2	129	+2.8	±5°	Passed
Wearable IMU	Shoulder abduction	31.7	34	+2.3	±5°	Passed
Wearable IMU	Shoulder abduction	64.2	66	+1.8	±5°	Passed
Wearable IMU	Shoulder abduction	107.3	109	+1.7	±5°	Passed
Wearable IMU	Knee flexion	27.7	30	+2.3	±5°	Passed
Wearable IMU	Knee flexion	48.1	50	+1.9	±5°	Passed
Wearable IMU	Knee flexion	73.3	75	+1.7	±5°	Passed

Table 11. Accuracy validation results of the rehabilitation glove and wearable IMU sensors under different rehabilitation movement conditions.

demonstrated high angular measurement accuracy across multiple rehabilitation movements, including shoulder flexion, shoulder abduction, elbow flexion, forearm supination, and knee flexion. The measured angular errors generally remained within approximately $\pm 3^\circ$, satisfying the predefined technical requirement of $\pm 5^\circ$. In particular, the maximum observed errors for shoulder flexion, forearm supination, elbow flexion, shoulder abduction, and knee flexion were 2.5° , 1.6° , 2.8° , 2.3° , and 2.3° , respectively.

All validation results satisfied the predefined technical requirements, indicating that the proposed wearable sensing system possesses good motion-measurement accuracy, stability, and reliability. These findings demonstrate that the system can accurately capture rehabilitation-related joint motion and finger-flexion information and can provide reliable motion data support for subsequent rehabilitation assessment, movement recognition, and intelligent rehabilitation analysis tasks.

Clinical and sample size validation. The baseline characteristics of the control and experimental groups are shown in Table 1. The t-test results indicated no significant differences between the two groups in terms of age, BMI, SBP, and DBP ($P > 0.05$). Although the mean stroke duration of the two groups was 61.20 and 46.22 days, respectively, the Wilcoxon test for non-normally distributed data showed that this difference was not statistically significant ($P > 0.05$). Chi-square tests revealed no significant differences between the two groups in terms of stroke type, HTN, ACD/MI, and gender distribution ($P > 0.05$). Additionally, Fisher’s exact test indicated no significant difference in HLP between the two groups ($P > 0.05$). These results demonstrate that all baseline characteristics showed no significant differences between the experimental group (60 participants) and the control group (60 participants), indicating that the initial conditions of both groups were balanced, without systematic bias⁶⁴. Therefore, in subsequent data analysis, the rehabilitation assessment data (including motion data and evaluation labels) from both groups can be considered to originate from comparable population distributions under randomized controlled trial conditions⁶⁵. From a statistical perspective, merging the data from both groups will not introduce systematic errors or compromise the reliability of the results. The combined dataset will be larger, improving data representativeness, enhancing the robustness and generalizability of model training, and providing a more comprehensive characterization of stroke patients’ motor function.

The experimental group received personalized rehabilitation training tailored to their condition, combined with conventional OT, while the control group underwent conventional OT combined with conventional PT. Both groups of hospitalized patients participated in rehabilitation training in the rehabilitation hall. By the end of the study, a total of 109 participants (91%) successfully completed the clinical trial and were included in the final statistical analysis, with a completion rate of 92% (55 participants) in the control group and 90% (54 participants) in the experimental group. Changes in FMA scores after two weeks of rehabilitation were analyzed using a t-test. The control group showed a mean total FMA score improvement of 11.98, with a 95% CI of 9.69–14.27 ($P = 0.005$), while the experimental group demonstrated a mean improvement of 17.56, with a 95% CI of 14.37–20.74 ($P = 0.01$). The t-test results indicated significant changes ($P < 0.05$) in the mean total FMA scores of both groups before and after the two-week training, confirming a notable improvement in patients’ motor function during the trial. For the downstream machine learning tasks investigated in this study, including feature extraction and pattern recognition, the pre-rehabilitation and post-rehabilitation assessment data were pooled to construct a larger dataset for model development. Since significant functional changes were observed after rehabilitation intervention, the pre- and post-rehabilitation motion data exhibited substantially different

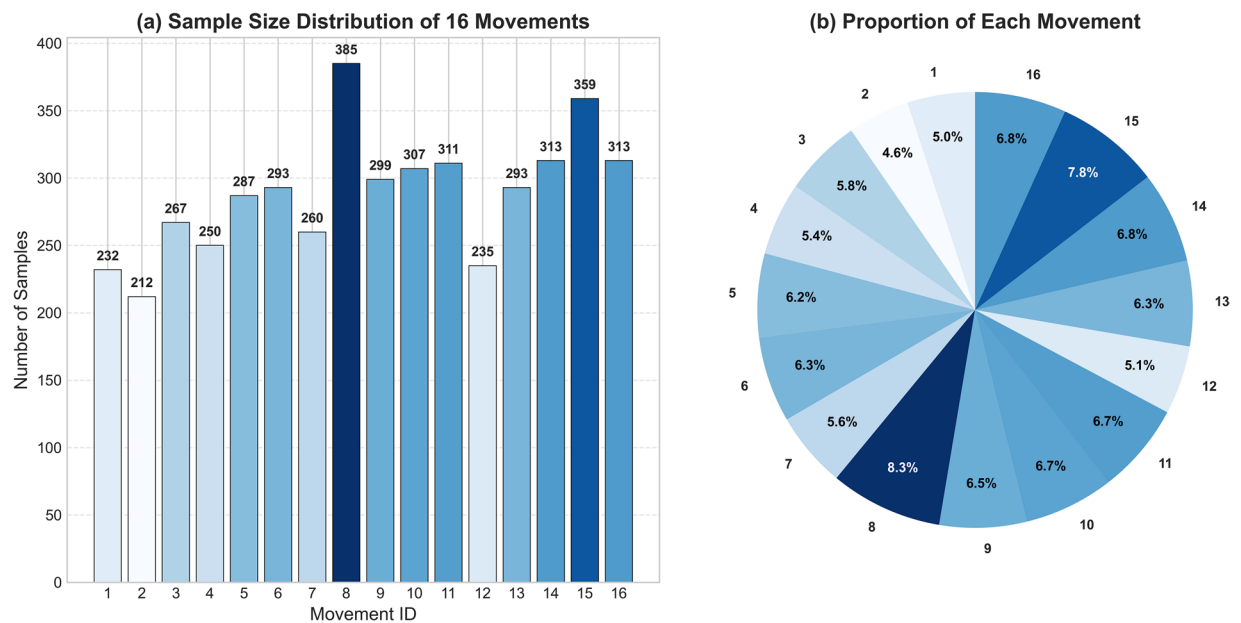


Fig. 5 Distribution of sample sizes for the 16 rehabilitation training movements. **(a)** Sample size distribution of the 16 movements. **(b)** Proportion of samples corresponding to each movement.

kinematic characteristics and assessment labels, which increased the diversity and representativeness of the dataset for rehabilitation modeling purposes.

In summary, statistical tests showed no significant differences in demographic and clinical parameters between the control and experimental groups at baseline, indicating that they were drawn from the same population and could therefore be combined. Furthermore, significant changes in motor function were observed in both groups before and after the intervention, suggesting that the two sets of assessment data (pre-rehabilitation and post-rehabilitation) can be treated as independent samples. In total, this resulted in 222 samples (baseline: 60 + 60; post-trial: 50 + 59; excluding 7 erroneous samples).

Effectiveness analysis of training movements. To analyze the effectiveness of the personalized exercise prescription on motor function recovery within the experimental group, this study compared the total FMA scores, upper limb scores, and lower limb scores (averaged scores) of all patients pre-rehabilitation and post-rehabilitation. Control group: Total FMA score increased from 41.11 (pre-rehabilitation) to 53.09 (post-rehabilitation), with an average increase of 11.98. Upper limb FMA score increased from 23.67 to 31.13, with an average increase of 7.45. Lower limb FMA score increased from 17.44 to 21.96, with an average increase of 4.52. Experimental group: Total FMA score increased from 43.28 (pre-rehabilitation) to 60.83 (post-rehabilitation), with an average increase of 17.56. Upper limb FMA score increased from 25.48 to 36.76, with an average increase of 11.28. Lower limb FMA score increased from 17.80 to 24.07, with an average increase of 6.28.

The minimal clinically important differences (MCID) for the between-group differences in total, upper limb, and lower limb FMA score improvements in post-stroke patients with hemiplegia were 2.22, 1.85, and 1.26 points, respectively⁶⁶. Compared with the control group, the experimental group achieved additional mean improvements of 5.58 points in total FMA score, 3.83 points in upper limb score, and 1.76 points in lower limb score. All of these between-group differences exceeded the corresponding MCID thresholds, indicating that the additional improvements achieved by the experimental group were clinically meaningful. These results suggest that patients who received personalized exercise prescription training experienced significantly greater enhancements in total, upper limb, and lower limb motor functions after 14 days of rehabilitation compared to those undergoing conventional therapy. This confirms the clinical effectiveness of the 16-movement training library used in the personalized exercise regimen, demonstrating its superiority over standard OT training in improving limb motor function. Therefore, the selection of these 16 movements for data collection is well-justified and meaningful, providing a strong foundation for subsequent model development.

Distribution of rehabilitation training data. A total of 4616 samples were collected for the 16 rehabilitation training actions. The sample size for each movement is shown in Fig. 5(a), and the proportion of each movement (in the total samples) is shown in Fig. 5(b). Due to individualized training prescriptions for each patient, the number of samples for each action varied, making some degree of data imbalance unavoidable. The figure illustrates the distribution of sample sizes across the 16 items. Statistical analysis showed that the number of samples per action ranged from 212 to 385, with an average of approximately 280 and a standard deviation of about 46. Overall, the sample sizes were relatively uniform, with no significant bias, indicating that the dataset is well-balanced across action categories and provides a reliable foundation for subsequent action recognition and deep learning modeling.

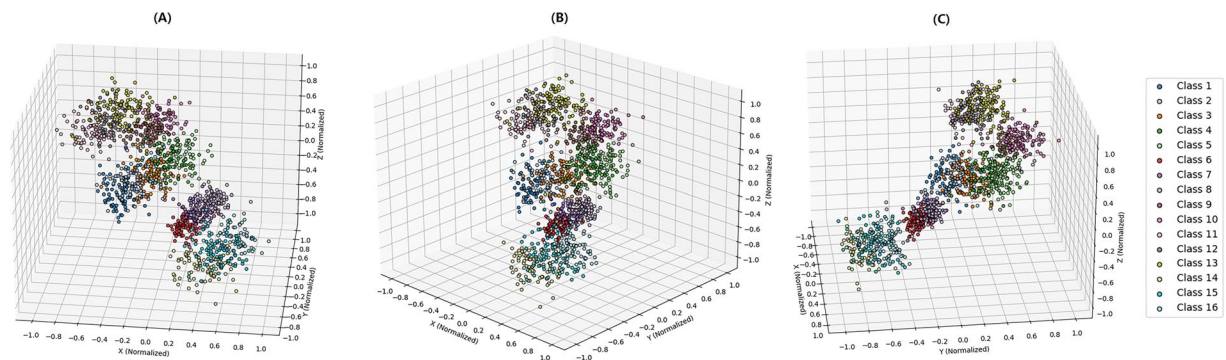


Fig. 6 Visualization of features for samples from 16 categories. (a), (b), and (c) represent three observational perspectives.

First, to examine the distribution characteristics of the data, we extracted the features of the samples, including amplitude, period, root mean square, and frequency. For each rehabilitation training sample (sample = $\{X_i, y_i\}$, where $X_i \in \mathbb{R}^{C \times N}$, $y_i \in \{0, 1, \dots, 16\}$), the feature extraction process is performed on each signal channel of X_i individually. Specifically, we extract predefined features from all C signal channels of X_i . This results in a total number of extracted features calculated as: C (number of channels in X_i) $\times$ number of feature types. For better visualization and understanding, we applied the K-means clustering algorithm⁶⁷, followed by Principal Component Analysis (PCA)⁶⁸ to reduce the dimensionality, mapping the high-dimensional data to a three-dimensional space. Figure 6 displays the three-dimensional visualization of the clustering results. From the figure, we can see that the data are grouped into five major categories and 16 smaller categories, with most samples being clearly distinguishable. The first major category includes categories 1–5, primarily involving upper limb movements (Arm/Fingers/Wrist/Shoulder), and these five categories are located close to each other. The second major category consists of categories 6–8, mainly related to fingers and wrists, with these three categories also positioned close to one another. The third major category includes categories 9–10, primarily related to shoulder activities. The fourth major category consists of categories 11–13, representing arm movements, while the fifth major category includes categories 14–16, mainly corresponding to lower limb movements.

However, we also observe some minor overlaps between categories. This may be due to similarities in certain movement patterns or overlapping sensor signal features. For example, categories 6–8 (fingers/wrists) overlap partially with categories 1–5 (related to the arms), and categories 9–10 (shoulder-related) may also overlap with certain arm movements (categories 11–13). This phenomenon indicates that although the clustering results generally reflect a good separation between categories, there are still some ambiguous areas due to the similarity of movement signals between certain categories. This overlap may be caused by improper feature selection, insufficient features, or data noise. To improve classification accuracy, we plan to optimize the feature selection process and experiment with more advanced algorithms, such as deep learning or network-based optimization methods, for more precise classification. These algorithms can learn deeper data features and automatically discover complex patterns within the data, thus improving classification accuracy and robustness. Overall, the clustering results demonstrate that the data distribution is reasonable and reliable, providing a solid foundation for subsequent pattern recognition and classification modeling.

Limitations. Although the REHAB dataset is intended to support intelligent home-based rehabilitation systems, all data were collected in supervised clinical environments. In real home-based rehabilitation scenarios, factors such as improper sensor wearing, environmental variability, lack of professional supervision, and inconsistent movement execution may introduce additional noise and variability. Therefore, although the current dataset provides a valuable foundation for developing intelligent rehabilitation algorithms, additional validation in real-world home environments is still necessary before large-scale practical deployment. Several potential sources of bias should also be acknowledged. First, all participants were recruited from two rehabilitation centers in China, which may limit the demographic and geographic diversity of the dataset. Second, although the sample size is relatively large compared with existing stroke rehabilitation datasets, the enrolled participants mainly consisted of patients who satisfied specific clinical inclusion criteria and were capable of completing rehabilitation assessments and training tasks. Therefore, the dataset may not fully represent patients with severe cognitive impairment, severe disability, or extremely limited motor function. Third, although standardized clinical procedures and therapist training were implemented throughout the study, some variability in therapist-guided movement execution and clinical scoring may still exist due to the subjective nature of rehabilitation assessment scales such as the Fugl-Meyer Assessment. These factors may introduce a certain degree of inter-operator variability into the dataset. Another limitation that should be acknowledged is the inconsistency between the channel configurations of the rehabilitation assessment dataset and the rehabilitation training dataset, as well as the fact that the standardized rehabilitation assessment labels were mainly derived from upper-limb movements. These differences were mainly related to the distinct acquisition objectives of the two scenarios. The rehabilitation assessment data retained richer signal channels to support fine-grained motor function evaluation, whereas the rehabilitation training data retained only representative channels to reduce storage burden during long-term recording. In addition, although both upper-limb and lower-limb rehabilitation training movements were included, the

current standardized rehabilitation assessment labels were primarily based on the upper-extremity section of the Fugl-Meyer Assessment (FMA-UE). The lower-limb assessment data have not yet been fully standardized and validated for public release because of data completeness and organization issues. In addition, the current dataset was manually segmented to extract relatively complete and stable motion cycles for model development and standardized data construction. In practical deployment scenarios, continuous sensor streams are more commonly processed using sliding-window-based automatic segmentation methods for real-time analysis.

Data availability

The dataset has been deposited in the Science Data Bank and is publicly available at <https://www.scidb.cn/en/s/YrUnie>. The dataset can also be accessed via its <https://doi.org/10.57760/sciencedb.37018>.

Code availability

All source code related to data preprocessing, dataset generation, and model training is available on GitHub: <https://github.com/zzzzz-123/REHAB>.

Received: 5 February 2026; Accepted: 29 June 2026;

Published online: 05 August 2026

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Acknowledgements

No other non-author contributions are acknowledged.

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Funding

The National Key Research and Development Program of China (2021YFF1201105).

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41597-026-07802-2>.

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